A Proof of the Snake Lemma (For R-Modules)

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1 Proof

Lemma 1 (The Snake Lemma)**.** *Let the diagram of R-modules*

$$\begin{tikzcd}
0 \arrow[r] & L_1 \arrow[r, "\alpha_1"] \arrow[d, "\lambda"] & M_1 \arrow[r, "\beta_1"] \arrow[d, "\mu"] & N_1 \arrow[r] \arrow[d, "\nu"] & 0 \\
0 \arrow[r] & L_0 \arrow[r, "\alpha_0"] & M_0 \arrow[r, "\beta_0"] & N_1 \arrow[r] & 0
\end{tikzcd}$$

commute, where rows are exact complexes. Then there exists an exact sequence

$$0 \longrightarrow \ker\lambda \longrightarrow \ker\mu \longrightarrow \ker\nu \xrightarrow{\delta} \operatorname{coker}\lambda \longrightarrow \operatorname{coker}\mu \longrightarrow \operatorname{coker}\nu \longrightarrow 0.$$

Proof. For convenience of notation, we will denote rows by $1_\bullet$ and $0_\bullet$, and columns by $L_\bullet$, $M_\bullet$, and $N_\bullet$.

Clearly, there exists the morphism $0 : 0 \to \ker\lambda$ with $\ker 0 = 0$ and $\operatorname{im} 0 = \{0\}$.

Since $1_\bullet$ is exact, $\ker\alpha_1 = \operatorname{im} 0 = \{0\}$. Consider the restriction $\alpha_1|_{\ker\lambda}$. Note that for any $a \in \ker\lambda$,

$$\begin{tikzcd}
a \arrow[r, "\alpha_1"] \arrow[d, "\lambda"'] & \alpha_1(a) \arrow[d, "\mu"] \\
0 \arrow[r] & 0
\end{tikzcd}$$

because the diagram commutes. We must then have $\operatorname{im}\alpha_1|_{\ker\lambda} \subseteq \ker\mu$. Therefore, the complex

$$0 \xrightarrow{0} \ker\lambda \xrightarrow{\alpha_1|_{\ker\lambda}} \ker\mu$$

is exact at $0, \ker\lambda$.

Now consider the restriction $\beta_1|_{\ker\mu} : \ker\mu \to N_1$. First, we will show $\operatorname{im}\beta_1 \subseteq \ker\nu$. Let $a \in \ker\mu$, and note that

$$\begin{tikzcd}
a \arrow[r, "\beta_1"] \arrow[d, "\mu"'] & \beta_1(a) \arrow[d, "\nu"] \\
0 \arrow[r] & 0
\end{tikzcd}$$

because the diagram commutes. Therefore, $\beta_1|_{\ker\mu} : \ker\mu \to \ker\nu$. Also note that because $1_\bullet$ is exact, we have $a \in \operatorname{im}\alpha_1 \iff a \in \ker\beta_1$. Restricting these sets to the subset $\ker\mu$ does not change this fact, so we have that the complex

$$0 \xrightarrow{0} \ker\lambda \xrightarrow{\alpha_1|_{\ker\lambda}} \ker\mu \xrightarrow{\beta_1|_{\ker\mu}} \ker\nu$$

is exact at $\ker\mu$.

Now we must show the morphism $\delta : \ker\nu \to \operatorname{coker}\lambda$ exists, which will be done by a diagram chase. Consider the diagram

$$
\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \ker\lambda \arrow[r] \arrow[d] & \ker\mu \arrow[r] \arrow[d] & \ker\nu \arrow[d] & \\
0 \arrow[r] & L_1 \arrow[r, "\alpha_1"] \arrow[d, "\lambda"] & M_1 \arrow[r, "\beta_1"] \arrow[d, "\mu"] & N_1 \arrow[r] \arrow[d, "\nu"] & 0 \\
0 \arrow[r] & L_0 \arrow[r, "\alpha_0"] \arrow[d] & M_0 \arrow[r, "\beta_0"] \arrow[d] & N_0 \arrow[r] \arrow[d] & 0 \\
 & \operatorname{coker}\lambda \arrow[r] \arrow[d] & \operatorname{coker}\mu \arrow[r] \arrow[d] & \operatorname{coker}\nu \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}, \tag{1}
$$

where all unexplained arrows are induced completely naturally. Let $a \in \ker\nu$, and consider the chase

$$
\begin{tikzcd}
 & 0 \arrow[d, dotted, no head] & 0 \arrow[d, dotted, no head] & 0 \arrow[d, dotted, no head] & \\
0 \arrow[r, dotted, no head] & {} \arrow[r, dotted, no head] \arrow[d, dotted, no head] & {} \arrow[r, dotted, no head] \arrow[d, dotted, no head] & a \arrow[d] & \\
0 \arrow[r, dotted, no head] & {} \arrow[r, dotted, no head] \arrow[d, dotted, no head] & c \arrow[r, "\beta_1"] \arrow[d, "\mu"] & b \arrow[r, dotted, no head] \arrow[d, dotted, no head, "\nu"] & 0 \\
0 \arrow[r, dotted, no head] & e \arrow[r, "\alpha_0"] \arrow[d] & d \arrow[r, dotted, no head, "\beta_0"'] \arrow[d, dotted, no head] & * \arrow[r, dotted, no head] \arrow[d, dotted, no head] & 0 \\
 & f \arrow[r, dotted, no head] \arrow[d, dotted, no head] & {} \arrow[r, dotted, no head] \arrow[d, dotted, no head] & {} \arrow[r, dotted, no head] \arrow[d, dotted, no head] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}. \tag{2}
$$

We will show this diagram induces a homomorphism $\delta : \ker\nu \to \operatorname{coker}\lambda$. The map $\ker\nu \hookrightarrow N_1$ is the inclusion map, the map $M_1 \to M_0$ is μ, and the map $L_0 \twoheadrightarrow \operatorname{coker}\lambda$ is the natural projection. The only maps that need sufficient explaining are the maps $N_1 \to M_1$ and $M_0 \to L_0$.

Given that $b = \iota(a)$ for $\iota : \ker\lambda \hookrightarrow N_1$ the inclusion map, we define c as the preimage $c \coloneqq \beta_1^{-1}(b)$. Such a c must exist, as β_1 must be epic for $1_\bullet$ to be exact by lemma 2. The question is whether this operation is well-defined; that is, different choices $c_1, c_2 \mapsto b$ must map to the same end result if we continue along the sequence. To show this, we must first define the map $L_0 \to M_0$.

This map is defined similarly. Since the diagram commutes, $\beta_0(d) = \nu(b)$. However, since $b \in \ker\nu$, we have $\beta_0(d) = 0$. It follows that $d \in \ker\beta$, and since $0_\bullet$ is exact, we have that $\operatorname{im}\alpha_0 = \ker\beta_0$, and there exists an $e \in L_0$ such that $\alpha_0(e) = d$. Additionally, note that since $0_\bullet$ is exact, α_0 is mono by lemma 2, so this e is unique. Therefore, we can define the map $M_0 \to L_0$ such that $d \mapsto \alpha_0^{-1}(d)$, and this is well-defined.

Now to show the map $b \mapsto c$ is well-defined. Let $c_1, c_2 \in M_1$ such that $\beta_1(c_1) = \beta_1(c_2) = b$. Since β_1 is a homomorphism, $\beta_1(c_1 - c_2) = 0$. Therefore, $c_1 - c_2 \in \ker \beta_1$. Since $1_\bullet$ is exact, $c_1 - c_2 \in \operatorname{im} \alpha_1$. Since $1_\bullet$ is exact, α_1 is mono by lemma 2, so there exists precisely one $g \in L_1$ such that $\alpha_1(g) = c_1 - c_2$. Consider the diagram

$$\begin{tikzcd}
0 \arrow[r, dash, dotted] & g \arrow[r, "\alpha_1"] \arrow[d, "\lambda"] & (c_1 - c_2) \arrow[r, "\beta_1"] \arrow[d, dash, dotted, "\mu"] & 0 \arrow[r, dash, dotted] \arrow[d, dash, dotted] & 0 \\
0 \arrow[r, dash, dotted] & \lambda(g) \arrow[r, dash, dotted] \arrow[d] & {} \arrow[r, dash, dotted] \arrow[d, dash, dotted] & {} & \\
& 0 \arrow[r, dash, dotted] & {} & &
\end{tikzcd}.$$

We need only explain $\lambda(g) \to 0$. It suffices to show columns are exact in diagram (1), since this would imply $\operatorname{im} \lambda = \ker \pi$ for $\pi : L_0 \to \operatorname{coker} \lambda$ the canonical projection. It suffices to show much less, actually, but I will just show this broader statement. Consider a complex of the form

$$0 \longrightarrow \ker \lambda \xhookrightarrow{\iota} L_1 \xrightarrow{\lambda} L_0 \xrightarrow{\pi} \operatorname{coker} \lambda \longrightarrow 0$$

Since $\iota : \ker \lambda \hookrightarrow L_1$ is injective by the universal property of kernels, we have that $\ker \iota = \{0\} = \operatorname{im} 0$, and the complex is exact at $\ker \lambda$. Note that $\ker \lambda = \iota(\ker \lambda) = \operatorname{im} \iota$, so the complex is exact at L_1. Now note that since $\operatorname{coker} \lambda \cong L_0 / \operatorname{im} \lambda$, we have that $\operatorname{im} \lambda \subseteq \ker \pi$. Additionally, $x \in \ker \pi$ implies that $\pi(x) = x + \operatorname{im} \lambda = \operatorname{im} \lambda$. By properties of cosets, it follows that $x \in \operatorname{im} \lambda$. Therefore, the complex is exact at L_0. Finally, recall the universal property of cokernels: the cokernel $\operatorname{coker} \lambda$ must be initial with respect to morphisms $\beta : L_0 \to Z$ such that the diagram

$$\begin{tikzcd}
L_1 \arrow[r, "\lambda"] \arrow[rr, bend left, "0"] & L_0 \arrow[r, "\beta"] \arrow[d, two heads, "\pi"] & Z \\
& \operatorname{coker} \lambda \arrow[ur, dotted, "\overline{\beta}"] &
\end{tikzcd}$$

commutes. Note that π is epic, so $\operatorname{im} \pi = \operatorname{coker} \lambda = \ker 0$, so the complex is exact at $\operatorname{coker} \lambda$. Rows are thus exact complexes, and we have that $\pi(\lambda(g)) = 0$ since $\operatorname{im} \lambda = \ker \pi$.

Now back to showing $b \mapsto c$ is well-defined. Suppose that $c_2 \mapsto d \mapsto e$. Since the diagram commutes, we must have $\mu \alpha_1(g) = \alpha_0 \lambda(g)$. We also have that $c_1 = c_2 + (c_1 - c_2) = c_2 + \alpha_1(g)$. Since $0_\bullet$ is epic, by lemma 2, α_0 is mono, and thus the only thing mapping to d is e, and thus $\mu(c_2) = \alpha_0(e)$ by commutativity. Now note that by commutativity of the diagram, $\alpha_0(e + \lambda(g)) = \mu(c_1) = \mu(c_2 + c_1 - c_2)$:

$$\begin{tikzcd}
& c_2 + c_1 - c_2 \arrow[d] \\
e + \lambda(g) \arrow[r] & d + \alpha_0(\lambda(g))
\end{tikzcd}$$

Since $\lambda(g)$ dies in $\operatorname{coker} \lambda$, we have that $c_1 \mapsto f + 0 = f$ in diagram (2). Therefore, c_1, c_2 eventually map to the same thing, and δ is well-defined overall as the map

$$a \xmapsto{\iota} b \xmapsto{\beta_1^{-1}} c \xmapsto{\mu} d \xmapsto{\alpha_0^{-1}} e \xmapsto{\pi} f.$$

Now, we must show the complex

$$0 \xrightarrow{0} \ker\lambda \xrightarrow{\alpha_1|_{\ker\lambda}} \ker\mu \xrightarrow{\beta_1|_{\ker\mu}} \ker\nu \xrightarrow{\delta} \operatorname{coker}\lambda$$

is exact at $\ker\nu$. First, we show $\operatorname{im}\beta_1|_{\ker\mu} \subseteq \ker\delta$. Consider the commutative diagram

$$\begin{tikzcd}
\ker\mu \arrow[r, "\beta_1"] \arrow[d, "\iota"] & \ker\nu \arrow[d, "\iota"] \\
M_1 \arrow[d, "\mu"] & N_1 \arrow[l, "\beta_1^{-1}"] \\
0 &
\end{tikzcd}$$

Notice that by the commutativity of this diagram, we have that

$$a \xmapsto{\beta_1} \beta_1(a) \xmapsto{\iota} \beta_1(a) \xmapsto{\beta_1^{-1}} a \xmapsto{\mu} 0 \xrightarrow{\pi\alpha_0^{-1}} 0$$

Therefore, $\operatorname{im}\beta_1 \subseteq \ker\delta$. Now we show the converse. ■

A Definitions and Results Used in the Proof

Lemma 2. *Let*

$$0 \longrightarrow M_2 \xrightarrow{d_1} M_1 \xrightarrow{d_0} M_0 \longrightarrow 0$$

be a short exact chain complex. Then d_1 is a monomorphism and d_0 is an epimorphism.

Definition 1 (Kernel)**.** The kernel $\ker\phi$ of an R-module homomorphism $\phi : R \to S$ is the R-module which is final with respect to mappings $\alpha : S \to Z$ such that $\phi\alpha = 0$. That is, $\ker\phi$ is the R-module such that

$$\begin{tikzcd}
R \arrow[r, "\phi"] \arrow[rr, bend left, "0"] \arrow[dr, dotted] & S \arrow[r, "\alpha"] & Z \\
& \ker\phi \arrow[u, hook] &
\end{tikzcd}$$

commutes.